

BACKGROUND

♣Video 1Ma: Set Operations; Union, Intersection, Complement, Cartesian Product

◇Problem [1Ma1] Let $A = \{-2, -1, 0, 1, 2\}$ and $B = \{0, 1, 2\}$.

Write down explicitly the following sets. Notation is important.

$$\begin{array}{ll} A \cup B & A \cap B^c \\ A \cap B & (A \cap B) \times B \\ \mathcal{P}(A \cap B) \cap B & \mathcal{P}(A \cap B) \times B \end{array}$$

◇Problem [1Ma2] Write a definition for the set $\mathbb{E}$ of even natural numbers in bracket notation. You may refer to $\mathbb{N}$ in your definition. Do not use ellipsis, "...".

◇Problem [1Ma3] Find three different functions with domain $\{a, b\} \times \{b, c\}$ and target $S = \{X \mid X \subseteq \{a, b, c\}\}$. How many different functions are there with this domain and target?

♣Video 1Mb: Subset, powerset, emptyset. Double Inclusion Method

◇Problem [1Mb1] Find three different subsets of $\{0, 1, 2, 3\}$.

How many different subsets are there? (Try to do answer *without* listing them all.) ◇Problem [1Mb2] Let $A = \{a, b, c, d, e, f\}$ and $B = \{c, f, g\}$.

Write down explicitly the following sets. Notation is important.

$$\mathcal{P}(B).$$

$$\mathcal{P}(A \cap B) \cup B.$$

$$\mathcal{P}(A \cap B) \times B.$$

◇Problem [1Mb3] Let X, Y and Z be sets. Pick one of the following identities and show it using the double inclusion method.

$$\begin{array}{l} X \cup (Y \cap Z) = (X \cup Z) \cap (Z \cup X) \\ (X \cup Z) \cap Y = (Z \cap Y) \cup (Y \cap X) \end{array}$$

♣Video 1Ta: Relations, Functions one-to-one onto

◇Problem [1Ta1] Let $X = \{a, b, c\}$ and $Y = \{a, c\}$.

Find a one-to-one function, call it f , with domain $\mathcal{P}(Y)$ and target $\mathcal{P}(X)$.

Find an onto function, call it g , with $g : \mathcal{P}(X) \longrightarrow \mathcal{P}(Y)$.

Show that g is not one-to-one.

◇Problem [1Ta2] Consider functions with domain $\mathbb{N}$ and target $\mathbb{Z}$.

- Define one which is one-to-one but not onto.

- Define one which is onto but not one-to-one.

◇Problem [1Ta3] Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(n) = n^2 + n + 1$.

Show explicitly that f is one-to-one.

Show explicitly that f not onto.

♣Video 1Tb: Equivalence Relations, Reflexive, Symmetric, Transitive

◇Problem [1Tb1] Let $A = \{s, t, u, v\}$. Draw the diagrams two different relations, R and R' , on A which are equivalence relations, but for which s is not related to u , and in which there are no equivalence classes of cardinality 2.

◇Problem [1Tb2] Let $B = \{0, 1, 2\}$. Define the diagram of a relation on B which is symmetric and transitive, but not reflexive.

◇Problem [1Tb3] Let T be the relation on $\mathbb{Z}$, defined by

$$T = \{(n, m) \in \mathbb{Z} \times \mathbb{Z} \mid n - m = 16k \text{ for some } k \in \mathbb{Z}\}.$$

Show T is an equivalence relation.

What is the equivalence class of 100?

Are there finitely many equivalence classes?

♣Video 1Ra: Diagonal Argument of countability

◇Problem [1Ra1] Let $f : A \rightarrow B$ be one-to-one.

Which of the following *must* be true? Give a word or two of explanation.

- a) $|A| \leq |B|$, d) $A \subset B$,
- b) $B \subseteq A$, e) $|A| \subseteq |B|$,
- c) $|B| = |A|$, f) $|A| < |A| + |B|$.

◇Problem [1Ra2] Let $f : A \rightarrow B$ be one-to-one.

Which of the following statements is true? For each, give a few words of explanation.

- a) If A is countable, then B is countable.
- b) If A is uncountable, the B is uncountable.
- c) There exists a function $g : B \rightarrow A$ which is one-to-one.
- d) Both A and B cannot be the set $\mathbb{N}$.

◇Problem [1Ra3] Let $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be defined by $f(n, m) = 2^n 3^m$.

Show that f is one-to-one.

Show that f is not onto.

What does this say about the countability of $\mathbb{N} \times \mathbb{N}$?

♣Video 1Rb: Countability of Constructions

◇Problem [1Rb1] Using the properties of countable sets, Label each of the following infinite sets as countable or not-countable.

Add a word or two of explanation.

- a) $\mathbb{Q} \times \mathcal{P}(\mathbb{Z})$
- b) $(\mathbb{Q} \times \mathbb{Z}) \cup (\mathbb{Q} \times \mathbb{Z})$
- c) $\mathcal{P}(\mathbb{Q}) \times \mathbb{Z}$
- d) $\bigcup_{i=1}^{\infty} \{(i, k) \mid k \geq i^2\}$

◇Problem [1Rb2] Using the properties of countable sets, Label each of the following infinite sets as finite countably infinite, uncountable.

Add a word or two of explanation.

a) $\mathcal{P}(\mathbb{R})$.

b) $\mathcal{P}(\mathcal{P}(\mathbb{N}))$.

d) $\bigcup_{i=1}^{\infty} \mathcal{P}^i(\{1, 2, 3\})$

◇Problem [1Rb3] Let $\mathcal{C}$ be the set of rooted binary trees. Is $\mathcal{C}$ finite, countable, or uncountable.

♣ Video 1Fa: Recursive Definition

◇Problem [1Fa1] Define the set L recursively as follows.

BASIS: $0 \in L$.

RECURSIVE STEP: If $x \in L$, then $x + 1 \in L$ and $x - 1 \in L$.

If $x \in L$, then $x/2 \in L$.

CLOSURE: Every element in L is obtained from the Basis after a finite number of applications of the recursive step.

Find L_0 and L_3 .

Find any rational number which is not in L , and explain why?

◇Problem [1Fa2] Let p_0 be a variable. Define a set E recursively by:

BASIS: $p_0 \in E$.

RECURSIVE STEP: If $p_i \in E$, then $p_{i+1} \in E$.

$x, y \in E$, then $\neg x$, $(x \vee y)$, and $(x \wedge y) \in E$.

CLOSURE: Every element in L is obtained from m the Basis after a finite number of applications of the recursive step.

Find E_0 , E_1 and E_2 .

◇Problem [1Fa3] We saw earlier that a function can be written as a relation, and a relation is a type of set.

Using this, write factorial function recursively.

♣ Video 1Fb: Introduction to induction

◇Problem [1Fb1] Suppose you have the sequence of numbers which starts

0, 1, 4, 10, 20, 35, 56, 84, 120, 165, 220. What would be your best guess for the next number?

◇Problem [1Fb2] Can you find a formula for the general value?

◇Problem [1Fb3] Can you prove your formula is correct?